

Frank–Wolfe Complexity Separation Between Frobenius and Operator-Norm Geometries

Abstract

We compare the standard smoothness–radius factors governing the Frank–Wolfe convergence bounds under the Frobenius and operator-norm geometries. For the Gaussian associative-memory softmax loss with Zipf probabilities

$$p_i = \frac{i^{-\alpha}}{H_{N,\alpha}}, \quad H_{N,\alpha} := \sum_{j=1}^N j^{-\alpha},$$

we study

$$\text{FW}_F(\varepsilon) := \beta_F R_F^2(\varepsilon), \quad \text{FW}_{\text{op}}(\varepsilon) := \beta_{\text{op}} R_{\text{op}}^2(\varepsilon).$$

We assume

$$N = d^\kappa, \quad 0 < \kappa < 2, \quad \alpha \geq 0$$

with κ and α fixed. In the accuracy range where the available Frobenius-radius lower bound is nontrivial, we prove a polynomial separation between the two Frank–Wolfe complexity factors. The dependence on the target loss ε cancels up to logarithmic factors, uniformly over the entire admissible accuracy range.

1 Setting and interpretation

Let

$$L(W) = \sum_{i=1}^N p_i \log \left(1 + \sum_{j \neq i} \exp((u_j - u_i)^\top W v_i) \right),$$

where the input and output embeddings are independent normalized Gaussian vectors in $\mathbb{R}^d$.

For $X \in \{F, \text{op}\}$, define

$$R_X(\varepsilon) := \inf \{ \|W\|_X : L(W) \leq \varepsilon \}.$$

Let β_X denote the global smoothness constant of L with respect to $\|\cdot\|_X$.

For a β_X -smooth objective over an X -norm ball of radius R , the Frank–Wolfe curvature constant is bounded, up to a universal constant, by

$$C_L \lesssim \beta_X R^2.$$

Consequently, the geometry-dependent factor in the standard Frank–Wolfe convergence estimate is

$$\text{FW}_X(\varepsilon) := \beta_X R_X^2(\varepsilon).$$

This note compares these standard smoothness–radius factors. It does not claim a minimax lower bound on the actual number of iterations of every possible algorithm.

Throughout, $\tilde{O}$, $\tilde{\Omega}$, and $\tilde{\Theta}$ suppress powers of

$$\log d, \quad \log N, \quad \log(1/\delta),$$

but do not suppress polynomial powers of d or N . Constants may depend on the fixed Zipf exponent α .

We assume

$$\delta^{-1} = d^{O(1)}.$$

The operator-radius construction requires

$$N \leq c_0 \frac{d^2}{\log d}.$$

Since $N = d^\kappa$ with fixed $\kappa < 2$,

$$\frac{N \log d}{d^2} = \frac{\log d}{d^{2-\kappa}} \longrightarrow 0,$$

so this condition holds for all sufficiently large d .

Theorem 1 (Main Complexity Separation). *Consider the Gaussian associative-memory softmax loss with Zipf probabilities $p_i \propto i^{-\alpha}$ for a fixed exponent $\alpha \geq 0$. Assume the number of facts is $N = d^\kappa$ for a fixed $0 < \kappa < 2$, and let the failure probability be $\delta^{-1} = d^{O(1)}$.*

There exist constants $c_\alpha > 0$ such that for any target accuracy ε in the range

$$0 < \varepsilon \leq c_\alpha \begin{cases} 1, & 0 \leq \alpha < 1, \\ (\log N)^{-1}, & \alpha = 1, \\ N^{1-\alpha}, & \alpha > 1, \end{cases}$$

the ratio of the Frank–Wolfe smoothness–radius factors satisfies, with probability at least $1 - \delta$,

$$\frac{\text{FW}_F(\varepsilon)}{\text{FW}_{\text{op}}(\varepsilon)} := \frac{\beta_F R_F^2(\varepsilon)}{\beta_{\text{op}} R_{\text{op}}^2(\varepsilon)} = \begin{cases} \tilde{\Omega}_\alpha(d^{\kappa\alpha}), & 0 < \kappa \leq 1, \quad 0 \leq \alpha < 1, \\ \tilde{\Omega}_\alpha(d^\kappa), & 0 < \kappa \leq 1, \quad \alpha \geq 1, \\ \tilde{\Omega}_\alpha(d^{(\kappa-1)(1-\alpha)} + d^\alpha), & 1 < \kappa < 2, \quad 0 \leq \alpha < 1, \\ \tilde{\Omega}_\alpha(d), & 1 < \kappa < 2, \quad \alpha \geq 1. \end{cases}$$

2 Input estimates

Set

$$m := \min\{d, N\}, \quad P_m := \sum_{i=1}^m p_i = \frac{H_{m,\alpha}}{H_{N,\alpha}}.$$

The smoothness estimates give, with high probability,

$$\beta_F \asymp \frac{1}{d} + p_1 = \frac{1}{d} + \frac{1}{H_{N,\alpha}}, \tag{1}$$

and

$$\beta_{\text{op}} \lesssim_\alpha (\log d)^2 \left(\frac{1}{d} + P_m \right). \tag{2}$$

The power $(\log d)^2$ in (2) is a convenient uniform upper bound. For fixed $\alpha > 0$, the available estimate can be sharpened by one logarithmic factor, but this does not affect any polynomial separation derived below.

The operator-radius construction gives, for every $0 < \varepsilon \leq 1$,

$$R_{\text{op}}^2(\varepsilon) \lesssim \left(1 + \frac{N}{d}\right) \log^2 \left(\frac{2N}{\varepsilon}\right). \quad (3)$$

For the Frobenius radius, introduce the natural probability scale

$$q_\alpha(N) := \begin{cases} 1, & 0 \leq \alpha < 1, \\ (\log N)^{-1}, & \alpha = 1, \\ N^{1-\alpha}, & \alpha > 1. \end{cases} \quad (4)$$

After changing constants depending only on α , the Frobenius lower bounds can be written uniformly as

$$R_F^2(\varepsilon) \gtrsim_\alpha \frac{N}{\log d} \log^2 \left(\frac{c_\alpha q_\alpha(N)}{\varepsilon} \right), \quad 0 < \varepsilon \leq c_\alpha q_\alpha(N). \quad (5)$$

More explicitly,

$$R_F^2(\varepsilon) \gtrsim_\alpha \begin{cases} \frac{N}{\log d} \log^2 \left(\frac{c_\alpha}{\varepsilon} \right), & 0 \leq \alpha < 1, \quad 0 < \varepsilon \leq c_\alpha, \\ \frac{N}{\log d} \log^2 \left(\frac{c}{\varepsilon \log N} \right), & \alpha = 1, \quad 0 < \varepsilon \leq \frac{c}{\log N}, \\ \frac{N}{\log d} \log^2 \left(\frac{c_\alpha}{\varepsilon N^{\alpha-1}} \right), & \alpha > 1, \quad 0 < \varepsilon \leq c_\alpha N^{1-\alpha}. \end{cases} \quad (6)$$

3 Why the accuracy dependence disappears in the ratio

The logarithms in (3) and (5) do not literally coincide. Nevertheless, their ratio is bounded below by an inverse polylogarithmic factor uniformly over the entire relevant accuracy range.

Lemma 1 (Uniform comparison of the accuracy logarithms). *After decreasing c_α if necessary, suppose*

$$0 < \varepsilon \leq c_\alpha q_\alpha(N)$$

and define

$$\ell_F(\varepsilon) := \log \left(\frac{c_\alpha q_\alpha(N)}{\varepsilon} \right), \quad \ell_{\text{op}}(\varepsilon) := \log \left(\frac{2N}{\varepsilon} \right).$$

Then

$$\frac{\ell_F^2(\varepsilon)}{\ell_{\text{op}}^2(\varepsilon)} \gtrsim_\alpha \frac{1}{(\log d)^2}.$$

In particular, no polynomial dependence on ε remains in the complexity ratio.

Proof. By shrinking c_α , we may assume

$$\ell_F(\varepsilon) \geq 1.$$

Moreover,

$$\begin{aligned}
\ell_{\text{op}}(\varepsilon) &= \log \left(\frac{2N}{\varepsilon} \right) \\
&= \log \left(\frac{c_\alpha q_\alpha(N)}{\varepsilon} \right) + \log \left(\frac{2N}{c_\alpha q_\alpha(N)} \right) \\
&= \ell_F(\varepsilon) + \log \left(\frac{2N}{c_\alpha q_\alpha(N)} \right).
\end{aligned} \tag{7}$$

For fixed α ,

$$\frac{N}{q_\alpha(N)} = \begin{cases} N, & 0 \leq \alpha < 1, \\ N \log N, & \alpha = 1, \\ N^\alpha, & \alpha > 1. \end{cases}$$

Since $N = d^\kappa$ and κ is fixed,

$$\log \left(\frac{2N}{c_\alpha q_\alpha(N)} \right) \lesssim_{\alpha, \kappa} \log d.$$

Using $\ell_F(\varepsilon) \geq 1$ in (7),

$$\ell_{\text{op}}(\varepsilon) \leq \ell_F(\varepsilon) + C_{\alpha, \kappa} \log d \leq (1 + C_{\alpha, \kappa} \log d) \ell_F(\varepsilon).$$

Therefore,

$$\frac{\ell_F^2(\varepsilon)}{\ell_{\text{op}}^2(\varepsilon)} \geq \frac{1}{(1 + C_{\alpha, \kappa} \log d)^2} \gtrsim_{\alpha, \kappa} \frac{1}{(\log d)^2}.$$

□

Remark 1. *The restriction*

$$\varepsilon \lesssim_\alpha q_\alpha(N)$$

is essential for the present Frobenius-radius lower bound. In particular, when $\alpha > 1$, the relevant accuracy range is

$$\varepsilon \lesssim_\alpha N^{1-\alpha}.$$

At larger accuracies, the population loss may ignore a block containing $\Theta(N)$ low-probability facts, so the current $K \asymp N$ Frobenius lower bound need not remain nontrivial.

4 A unified Frank–Wolfe separation formula

Define

$$\text{Sep}(\varepsilon) := \frac{\text{FW}_F(\varepsilon)}{\text{FW}_{\text{op}}(\varepsilon)} = \frac{\beta_F R_F^2(\varepsilon)}{\beta_{\text{op}} R_{\text{op}}^2(\varepsilon)}.$$

Theorem 2 (Unified complexity separation). *Let $\alpha \geq 0$ and $N = d^\kappa$ with fixed $0 < \kappa < 2$. For*

$$0 < \varepsilon \leq c_\alpha q_\alpha(N),$$

with high probability,

$$\text{Sep}(\varepsilon) \gtrsim_\alpha \frac{1}{(\log d)^5} \min\{N, d\} \frac{d^{-1} + p_1}{d^{-1} + P_{\min\{N, d\}}}. \tag{8}$$

Equivalently,

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha \left(\min\{N, d\} \frac{d^{-1} + p_1}{d^{-1} + P_{\min\{N, d\}}} \right)}. \tag{9}$$

Proof. Combining (1) and (5),

$$\text{FW}_F(\varepsilon) \gtrsim_\alpha \left(\frac{1}{d} + p_1 \right) \frac{N}{\log d} \ell_F^2(\varepsilon).$$

Combining (2) and (3),

$$\text{FW}_{\text{op}}(\varepsilon) \lesssim_\alpha (\log d)^2 \left(\frac{1}{d} + P_m \right) \left(1 + \frac{N}{d} \right) \ell_{\text{op}}^2(\varepsilon).$$

Thus

$$\text{Sep}(\varepsilon) \gtrsim_\alpha \frac{N}{1 + N/d} \frac{d^{-1} + p_1}{d^{-1} + P_m} \frac{1}{(\log d)^3} \frac{\ell_F^2(\varepsilon)}{\ell_{\text{op}}^2(\varepsilon)}. \quad (10)$$

Lemma 1 gives

$$\frac{\ell_F^2(\varepsilon)}{\ell_{\text{op}}^2(\varepsilon)} \gtrsim_\alpha \frac{1}{(\log d)^2}.$$

Finally,

$$\frac{N}{1 + N/d} = \frac{Nd}{N + d} \asymp \min\{N, d\}.$$

Substitution into (10) proves (8) and (9). $\square$

5 Evaluation of the Zipf quantities

The standard generalized-harmonic-number estimates are

$$H_{n,\alpha} \asymp_\alpha \begin{cases} n^{1-\alpha}, & 0 \leq \alpha < 1, \\ \log n, & \alpha = 1, \\ 1, & \alpha > 1. \end{cases}$$

Consequently,

$$p_1 = \frac{1}{H_{N,\alpha}} \asymp_\alpha \begin{cases} N^{\alpha-1}, & 0 \leq \alpha < 1, \\ (\log N)^{-1}, & \alpha = 1, \\ 1, & \alpha > 1, \end{cases} \quad (11)$$

and

$$P_m = \frac{H_{m,\alpha}}{H_{N,\alpha}} \asymp_\alpha \begin{cases} \left(\frac{m}{N} \right)^{1-\alpha}, & 0 \leq \alpha < 1, \\ \frac{\log m}{\log N}, & \alpha = 1, \\ 1, & \alpha > 1. \end{cases} \quad (12)$$

We now evaluate (9) in all regimes.

6 Regime I: $N \leq d$

Suppose

$$0 < \kappa \leq 1, \quad N = d^\kappa \leq d.$$

Then

$$m = N, \quad P_m = P_N = 1.$$

Therefore,

$$\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha \left(N \left(\frac{1}{d} + p_1 \right) \right).$$

6.1 $0 \leq \alpha < 1$

In this regime,

$$p_1 \asymp_\alpha N^{\alpha-1}.$$

Since $N \leq d$ and $0 < 1 - \alpha \leq 1$,

$$N^{\alpha-1} = \frac{1}{N^{1-\alpha}} \geq \frac{1}{d}.$$

Hence

$$\frac{1}{d} + p_1 \asymp_\alpha N^{\alpha-1},$$

and thus

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha(d^{\kappa\alpha}), \quad 0 \leq \alpha < 1, \quad 0 < \kappa \leq 1.} \quad (13)$$

At $\alpha = 0$, this gives only

$$\text{Sep}(\varepsilon) = \tilde{\Omega}(1),$$

so the present estimates do not yield a polynomial separation in the uniform-frequency regime when $N \leq d$.

For every fixed $\alpha > 0$, however,

$$d^{\kappa\alpha} \rightarrow \infty,$$

and hence there is a polynomial advantage for the operator geometry.

6.2 $\alpha = 1$

Here

$$p_1 \asymp \frac{1}{\log N},$$

which dominates $1/d$. Therefore,

$$\text{Sep}(\varepsilon) = \tilde{\Omega} \left(\frac{N}{\log N} \right) = \tilde{\Omega}(N).$$

Since $N = d^\kappa$,

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega}(d^\kappa), \quad \alpha = 1, \quad 0 < \kappa \leq 1.} \quad (14)$$

6.3 $\alpha > 1$

In this case,

$$p_1 \asymp_\alpha 1,$$

so

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha(d^\kappa), \quad \alpha > 1, \quad 0 < \kappa \leq 1.} \quad (15)$$

Combining all three cases,

$$\boxed{\text{Sep}(\varepsilon) = \begin{cases} \tilde{\Omega}_\alpha(d^{\kappa\alpha}), & 0 \leq \alpha < 1, \\ \tilde{\Omega}_\alpha(d^\kappa), & \alpha \geq 1, \end{cases} \quad 0 < \kappa \leq 1.} \quad (16)$$

Equivalently,

$$\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha\left(d^{\kappa \min\{\alpha, 1\}}\right), \quad 0 < \kappa \leq 1.$$

7 Regime II: $d < N < d^2$

Suppose

$$1 < \kappa < 2, \quad N = d^\kappa.$$

Then

$$m = d.$$

7.1 $0 \leq \alpha < 1$

We have

$$p_1 \asymp_\alpha N^{\alpha-1}, \quad P_d \asymp_\alpha \left(\frac{d}{N}\right)^{1-\alpha}.$$

Moreover, because $d < N < d^2$,

$$\left(\frac{d}{N}\right)^{1-\alpha} \geq \frac{d}{N} > \frac{1}{d}.$$

Therefore,

$$\frac{1}{d} + P_d \asymp_\alpha \left(\frac{d}{N}\right)^{1-\alpha}.$$

The master formula gives

$$\begin{aligned} \text{Sep}(\varepsilon) &= \tilde{\Omega}_\alpha \left[d \frac{d^{-1} + N^{\alpha-1}}{(d/N)^{1-\alpha}} \right] \\ &= \tilde{\Omega}_\alpha \left[\frac{1}{(d/N)^{1-\alpha}} + \frac{dN^{\alpha-1}}{(d/N)^{1-\alpha}} \right] \\ &= \tilde{\Omega}_\alpha \left[\left(\frac{N}{d}\right)^{1-\alpha} + d^\alpha \right]. \end{aligned} \quad (17)$$

Thus

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha \left[\left(\frac{N}{d}\right)^{1-\alpha} + d^\alpha \right], \quad d < N < d^2, \quad 0 \leq \alpha < 1.} \quad (18)$$

Using $N = d^\kappa$,

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha \left[d^{(\kappa-1)(1-\alpha)} + d^\alpha \right] = \tilde{\Omega}_\alpha \left(d^{\max\{(\kappa-1)(1-\alpha), \alpha\}} \right).} \quad (19)$$

The two powers are equal when

$$(\kappa - 1)(1 - \alpha) = \alpha.$$

Solving gives the phase boundary

$$\alpha_\star = 1 - \frac{1}{\kappa}. \quad (20)$$

Therefore,

$$\boxed{\text{Sep}(\varepsilon) = \begin{cases} \tilde{\Omega}_\alpha \left(d^{(\kappa-1)(1-\alpha)} \right), & 0 \leq \alpha \leq 1 - \frac{1}{\kappa}, \\ \tilde{\Omega}_\alpha (d^\alpha), & 1 - \frac{1}{\kappa} \leq \alpha < 1. \end{cases}} \quad (21)$$

The two expressions agree at $\alpha = 1 - 1/\kappa$.

The origin of the two terms in (17) is worth emphasizing:

$$\left(\frac{N}{d} \right)^{1-\alpha}$$

comes from the diffuse d^{-1} component of β_F , whereas

$$d^\alpha$$

comes from the largest-atom component $p_1 \asymp N^{\alpha-1}$.

In particular, at $\alpha = 0$,

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega} \left(d^{\kappa-1} \right).}$$

Thus, even for uniform frequencies, a polynomial separation emerges as soon as $N > d$.

7.2 $\alpha = 1$

For $N = d^\kappa$ with $\kappa > 1$,

$$p_1 \asymp \frac{1}{\log N}, \quad P_d \asymp \frac{\log d}{\log N} = \frac{1}{\kappa} \asymp 1.$$

Hence

$$\text{Sep}(\varepsilon) = \tilde{\Omega} \left(d^{\frac{1/\log N}{1}} \right) = \tilde{\Omega} \left(\frac{d}{\log N} \right).$$

Since $\log N = \kappa \log d$,

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega}(d), \quad \alpha = 1, \quad 1 < \kappa < 2.} \quad (22)$$

7.3 $\alpha > 1$

For fixed $\alpha > 1$,

$$p_1 \asymp_\alpha 1, \quad P_d \asymp_\alpha 1.$$

Therefore,

$$\boxed{\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha(d), \quad \alpha > 1, \quad 1 < \kappa < 2.} \quad (23)$$

Combining the three cases,

$$\boxed{\text{Sep}(\varepsilon) = \begin{cases} \tilde{\Omega}_\alpha [d^{(\kappa-1)(1-\alpha)} + d^\alpha], & 0 \leq \alpha < 1, \\ \tilde{\Omega}_\alpha(d), & \alpha \geq 1, \end{cases} \quad 1 < \kappa < 2.} \quad (24)$$

Equivalently, the polynomial exponent can be written compactly as

$$\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha \left(d^{s(\kappa, \alpha)} \right),$$

where

$$s(\kappa, \alpha) = \begin{cases} \kappa \min\{\alpha, 1\}, & 0 < \kappa \leq 1, \\ \min\{1, \max\{\alpha, (\kappa - 1)(1 - \alpha)\}\}, & 1 < \kappa < 2. \end{cases} \quad (25)$$

8 Equivalent comparison of the two complexity factors

The preceding results can also be understood by comparing the polynomial parts of the numerator and denominator separately.

Case $N \leq d$

Since

$$\beta_{\text{op}} = \Theta(1), \quad R_{\text{op}}^2(\varepsilon) \lesssim \log^2 \left(\frac{2N}{\varepsilon} \right),$$

the operator complexity factor has no polynomial dependence on d or N :

$$\text{FW}_{\text{op}}(\varepsilon) \leq \text{polylog}(d, N, 1/\varepsilon).$$

On the other hand,

$$\text{FW}_F(\varepsilon) \gtrsim_\alpha \frac{N}{\log d} \left(\frac{1}{d} + p_1 \right) \ell_F^2(\varepsilon).$$

Its polynomial part is therefore

$$\begin{cases} N^\alpha, & 0 \leq \alpha < 1, \\ N, & \alpha \geq 1, \end{cases}$$

up to logarithmic factors.

Case $d < N < d^2$

For $0 \leq \alpha < 1$,

$$\beta_{\text{op}} = \tilde{O}_\alpha \left(\left(\frac{d}{N} \right)^{1-\alpha} \right), \quad R_{\text{op}}^2(\varepsilon) \lesssim \frac{N}{d} \log^2 \left(\frac{2N}{\varepsilon} \right).$$

Thus the polynomial part of the operator complexity factor is

$$\text{FW}_{\text{op}}(\varepsilon) = \tilde{O}_\alpha \left(\left(\frac{N}{d} \right)^\alpha \right),$$

whereas

$$\text{FW}_F(\varepsilon) = \tilde{\Omega}_\alpha \left(\frac{N}{d} + N^\alpha \right)$$

after matching the accuracy logarithms. Consequently,

$$\begin{aligned} \frac{\text{FW}_F(\varepsilon)}{\text{FW}_{\text{op}}(\varepsilon)} &= \tilde{\Omega}_\alpha \left(\frac{N/d + N^\alpha}{(N/d)^\alpha} \right) \\ &= \tilde{\Omega}_\alpha \left[\left(\frac{N}{d} \right)^{1-\alpha} + d^\alpha \right]. \end{aligned}$$

For $\alpha \geq 1$,

$$\beta_{\text{op}} = \Theta_\alpha(1), \quad R_{\text{op}}^2(\varepsilon) \lesssim \frac{N}{d} \log^2 \left(\frac{2N}{\varepsilon} \right),$$

while the Frobenius polynomial factor is of order N . The resulting separation is therefore of order d , up to polylogarithmic factors.

9 Final conclusion

For all fixed $\alpha \geq 0$ and fixed $0 < \kappa < 2$, in the accuracy range

$$0 < \varepsilon \lesssim_\alpha \begin{cases} 1, & 0 \leq \alpha < 1, \\ (\log N)^{-1}, & \alpha = 1, \\ N^{1-\alpha}, & \alpha > 1, \end{cases}$$

the Frank–Wolfe smoothness–radius factors satisfy

$$\frac{\beta_F R_F^2(\varepsilon)}{\beta_{\text{op}} R_{\text{op}}^2(\varepsilon)} = \begin{cases} \tilde{\Omega}_\alpha(d^{\kappa\alpha}), & 0 < \kappa \leq 1, \quad 0 \leq \alpha < 1, \\ \tilde{\Omega}_\alpha(d^\kappa), & 0 < \kappa \leq 1, \quad \alpha \geq 1, \\ \tilde{\Omega}_\alpha(d^{(\kappa-1)(1-\alpha)} + d^\alpha), & 1 < \kappa < 2, \quad 0 \leq \alpha < 1, \\ \tilde{\Omega}_\alpha(d), & 1 < \kappa < 2, \quad \alpha \geq 1. \end{cases} \quad (26)$$

For $1 < \kappa < 2$ and $0 \leq \alpha < 1$, the more explicit phase diagram is

$$\frac{\beta_F R_F^2(\varepsilon)}{\beta_{\text{op}} R_{\text{op}}^2(\varepsilon)} = \begin{cases} \tilde{\Omega}_\alpha(d^{(\kappa-1)(1-\alpha)}), & 0 \leq \alpha \leq 1 - \frac{1}{\kappa}, \\ \tilde{\Omega}_\alpha(d^\alpha), & 1 - \frac{1}{\kappa} \leq \alpha < 1. \end{cases}$$

Thus, except for the corner

$$\alpha = 0, \quad N \leq d,$$

the available bounds yield a polynomial separation in favor of the operator-norm geometry. The maximal separation is linear in the ambient dimension:

$$\text{Sep}(\varepsilon) = \tilde{\Omega}_\alpha(d)$$

throughout the heavy-head regime $\alpha \geq 1$ when $N > d$.